$$
\begin{aligned}
x_p &= r_p \cos\theta_p + \epsilon_{px} + \epsilon_{fx} + w_x \\
y_p &= r_p \sin\theta_p + \epsilon_{py} + \epsilon_{fy} + w_y \\
\widetilde{h}_p &= h(x_p, y_p) + \epsilon_{pz} + \epsilon_{fz} \\
&\quad + w_z + \epsilon_{\text{outlier}}
\end{aligned}
$$